

Section : Biology

Q.1 Scientific term for a situation where a single gene can exhibit multiple phenotypic expression is:

1. Polygenic inheritance
2. Pleiotropy
3. Co-dominance
4. Phylogeny

Options 1. 1

2. 2
3. 3
4. 4

Q.2 Which of the following statements are correct about restriction enzymes?

- (A). Each restriction endonuclease recognizes a specific palindromic nucleotide sequence in the DNA.
- (B). Endonuclease removes nucleotides from the terminal end of the DNA.
- (C). The first discovered restriction endonuclease is *Hind II*.
- (D). Restriction enzymes belong to a larger class of enzyme polymerase.

Choose the correct answer from the options given below:

1. (B) and (D) only
2. (A), and (C) only
3. (B), (C) and (D)
4. (A), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.3 Which one of the following revolutions succeeded in tripling the food supply?

1. Blue Revolution
2. Green Revolution
3. Grey Revolution
4. White Revolution

Options 1. 1

2. 2
3. 3
4. 4

Q.4 The spermatids are transformed into spermatozoa by the process called as -

1. Spermatocytogenesis
2. Spermatidogenesis
3. Spermiation
4. Spermiogenesis

Options 1. 1

2. 2
3. 3
4. 4

Q.5 The structures present in human sperm are-

- (A). Nucleus
- (B). Zona pellucida
- (C). Acrosome
- (D). Mitochondria

Choose the correct answer from the options given below:

1. (A) and (C) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (A), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.6

Which one of the following contributes the maximum percentage to the biodiversity of a vertebrate group?

1. Mammals
2. Fishes
3. Birds
4. Reptiles

Options 1. 1

2. 2

3. 3

4. 4

Q.7 Which one of the following statements explains autogamy?

1. Pollination is achieved within the same flower.
2. Transfer of pollens from one flower to another.
3. Transfer of pollens from the anther to the stigma of another flower of the same plant.
4. Transfer of pollens from stigma to anther of a different plant.

Options 1. 1

2. 2

3. 3

4. 4

Q.8 In the majority of flowering plants, a certain pattern is followed in megaspore mother cell (MMC) undergoing meiosis and producing megaspores. How many functional megaspores would be produced with 32 MMC?

1. 96
2. 32
3. 128
4. 62

Options 1. 1

2. 2

3. 3

4. 4

Q.9 To produce in large quantities, the development of bioreactors, where large volumes _____ of culture can be processed, is required.

1. 1-10 litres
2. 10-100 litres
3. 100-1000 litres
4. 10-20 litres

Options 1. 1

2. 2

3. 3

4. 4

Q.10 How much energy will be converted by the primary producer to net primary productivity from 20,000 J of available sunlight energy?

1. 200 J
2. 2000 J
3. 10,000 J
4. 20 kJ

Options 1. 1

2. 2

3. 3

4. 4

Q.11 Match List-I with List-II

List-I**Terms**

- (A). Amniocentesis
(B). MMR
(C). Periodic abstinence
(D). Vasectomy

List-II**Features**

- (I). Maternal mortality rate
(II). Surgical method of contraception
(III). Process to test genetic disorders in foetus
(IV). Natural method of contraception

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (III), (B) - (IV), (C) - (I), (D) - (III)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (I)
4. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.12 Which approach is not applied for *ex situ* conservation?

1. Zoological parks
2. Botanical gardens
3. National parks
4. Wildlife safari parks

Options 1. 1

2. 2
3. 3
4. 4

Q.13 If 4 individuals in a laboratory population of 40 fruitflies died during a specified time interval, say a week, the death rate in the population during that period is -

1. 10 individual per fruitfly per week
2. 0.1 individual per fruitfly per week
3. 1 individual per fruit-fly per week
4. 4 individual per fruitfly per week

Options 1. 1

2. 2
3. 3
4. 4

Q.14 Match List-I with List-II

List-I**Terms related to RNA**

- (A). Capping
(B). Polyadenylation
(C). RNA polymerase I
(D). Splicing

List-II**Features**

- (I). Transcribes rRNAs
(II). Removal of introns
(III). 5' end-methyl guanosine triphosphate addition
(IV). Addition of 200-300 adenylate residues

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.15 Identify the correct facts about downstream processing.

- (A) The processes include separation and purification.
(B) The product has to be formulated with suitable preservatives.

(C) The prepared formulation does not require clinical trials as in case of drugs.

(D) Strict quality control testing for each product is also required.

Choose the correct answer from the options given below:

1. (A), (B) and (C) only
2. (A), (B) and (D) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.16 The *cryIIAb* gene codes the protein for the specific insect group -

1. Cotton boll worm
2. Corn borer
3. *Drosophila*
4. Mosquito

Options 1. 1

2. 2

3. 3

4. 4

Q.17 Members of a population are arranged as per their age group. Select the expanding population from the following situations.

1. Reproductive age > Post reproductive age > Pre-reproductive age
2. Post reproductive age > Pre-reproductive age > Reproductive age
3. Pre-reproductive age > Reproductive age > Post reproductive age
4. Post reproductive age < Pre-reproductive age = Reproductive age

Options 1. 1

2. 2

3. 3

4. 4

Q.18 Progestasert and LNG-20 are examples of:

1. Copper releasing IUDs
2. Hormone releasing IUDs
3. Contraceptive pills
4. Non-medicated IUDs

Options 1. 1

2. 2

3. 3

4. 4

Q.19 The broadly utilitarian argument says that -

1. Biodiversity plays no role in many ecosystem services that nature provides.
2. Biodiversity plays a major role in many ecosystem services that nature provides.
3. Biodiversity plays a minimum role in many ecosystem services that nature provides.
4. Conserving biodiversity is not obvious.

Options 1. 1

2. 2

3. 3

4. 4

Q.20 The direction of transport of sperm in human is-

- (A). Vas deferens
- (B). Vasa efferentia
- (C). Epididymis
- (D). Ejaculatory duct

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)

2. (A), (C), (B), (D)
3. (B), (A), (D), (C)
4. (B), (C), (A), (D)

Options 1. 1

2. 2
3. 3
4. 4

Q.21 Which is not the salient feature of human genome project-

1. Less than 2 per cent of the genome codes for proteins.
2. Repeated sequences make up very small portion of the human genome.
3. The functions are unknown for over 50 per cent of the discovered genes.
4. The average gene consists of 3000 bases, but sizes vary greatly, with the largest known human gene being dystrophin at 2.4 million bases.

Options 1. 1

2. 2
3. 3
4. 4

Q.22 Arrange the correct sequence of steps for gel electrophoresis.

- (A). Load this DNA into the agarose gel matrix
 - (B). Cutting the DNA fragments by using a restriction enzyme
 - (C). Use ethidium bromide for DNA staining
 - (D). Separated bands of DNA are cut out from agarose gel
- Choose the correct answer from the options given below:

1. (B), (A), (C), (D)
2. (A), (C), (B), (D)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.23 Match List-I with List-II

List-I	List-II
Terms	Features
(A). Humus	(I). Photosynthetically active radiation
(B). PAR	(II). Biomass at a particular time
(C). Trophic level	(III). Dark-colored amorphous substance
(D). Standing crop	(IV). Nutrition based-specific place in the food chain

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)
2. (A) - (III), (B) - (I), (C) - (II), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.24 The transfer of the ovum from a donor to the fallopian tube of another female is known as:

1. ZIFT
2. IUT
3. *In-vivo* fertilization
4. GIFT

Options 1. 1

2. 2
3. 3

Q.25 Arrange the steps in correct order for somatic hybridization.

- (A). Isolated protoplasm are fused together
- (B). Isolate the protoplasts from a single cell of different plants
- (C). Digest the cell wall to isolate the protoplasts
- (D). Hybrid protoplasts grown to form new plants

Choose the correct answer from the options given below:

- 1. (A), (B), (C), (D)
- 2. (B), (C), (A), (D)
- 3. (B), (A), (D), (C)
- 4. (C), (B), (D), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.26 Factors that do not affect Hardy-Weinberg equilibrium are:

- (A). Genetic drift
- (B). Mitosis
- (C). Genetic recombination and natural selection
- (D). Pleiotropy

Choose the *correct* answer from the options given below:

- 1. (A), (B) and (D) only
- 2. (A), (B) and (C) only
- 3. (B) and (D) only
- 4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.27 Match List-I with List-II

List-I

List-II

Structures

Features

- (A). Infundibulum (I). Helps in collection of the ovum after ovulation
- (B). Zona pellucida (II). Finger-like structures above labia minora
- (C). Fimbriae (III). Membrane of secondary oocyte
- (D). Clitoris (IV). Funnel-shaped structure near ovary

Choose the *correct* answer from the options given below:

- 1. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)
- 2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
- 3. (A) - (I), (B) - (III), (C) - (IV), (D) - (II)
- 4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.28 Arrange the following plants as per the correct order of their evolution.

- (A). Psilophyton
- (B). Tracheophyte ancestors
- (C). Rhynia-type plants
- (D). Chlorophyte ancestors

Choose the *correct* answer from the options given below:

- 1. (A), (B), (C), (D)
- 2. (A), (C), (B), (D)
- 3. (D), (A), (B), (C)
- 4. (D), (B), (C), (A)

Options 1. 1

2. 2

3. 3

Q.29 The position of antipodals in mature embryo sac in angiosperm plant is -

1. At chalazal end
2. At micropylar end
3. In central part
4. In synergid

Options 1. 1

2. 2
3. 3
4. 4

Q.30 Match List-I with List-II

List-I	List-II
Enzymes	Used for
(A). Protease	(I). RNA
(B). Chitinase	(II). Protein
(C). Lysozyme	(III). Fungus
(D). Ribonuclease	(IV). Bacterial cell

Choose the *correct* answer from the options given below:

1. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.31 The first human-like being the hominid was called as -

1. *Homo sapiens*
2. *Homo erectus*
3. *Homo habilis*
4. Neanderthal man

Options 1. 1

2. 2
3. 3
4. 4

Q.32 Which one of the following codons performs a dual function?

1. UAG
2. AUG
3. UUU
4. AUA

Options 1. 1

2. 2
3. 3
4. 4

Q.33 The first _____ was given in 1990 to a 4-year old girl with adenosine deaminase (ADA) deficiency.

1. Chemotherapy
2. Radioactive therapy
3. Clinical gene therapy
4. Stem cell technology

Options 1. 1

2. 2
3. 3
4. 4

Q.34 Standing crop is defined as:

1. The mass of living material at a trophic level at a particular time
2. Biomass per kilogram
3. Crop of any plant in the field
4. The mass of dead and living organisms at a trophic level at a particular time

Options 1. 1

2. 2
3. 3
4. 4

Q.35 An insect ingests the inactive toxin, it is converted into an active form of toxin due to -

1. Acidic pH of the gut
2. Alkaline pH of the gut
3. Neutral pH of the gut
4. Low pH of the gut

Options 1. 1

2. 2
3. 3
4. 4

Q.36 Which one is not the example of an invasive alien species?

1. *Eicchornia*
2. *Lantana*
3. Quagga
4. *Clarias gariepinus*

Options 1. 1

2. 2
3. 3
4. 4

Q.37 Which one of the following regions is called as "lungs of the planet"?

1. Lake Victoria in East Africa
2. Amazon rain forest
3. Western Ghats and Sri Lanka
4. Himalaya and Indo-Burma regions

Options 1. 1

2. 2
3. 3
4. 4

Q.38 The DNA fingerprinting technique is based on -

1. Gene expression
2. Human Genome Project
3. Polymorphism
4. Heterochromatin

Options 1. 1

2. 2
3. 3
4. 4

Q.39 Identify the correct statements about test cross.

- (A). It can predict the genotype of the test organism
(B). It can distinguish homozygous and heterozygous among test organisms
(C). It is another name for back cross
(D). It is the law of dominance

Choose the *correct* answer from the options given below:

1. (A), (B) and (D) only
2. (A) and (B) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.40 The stages of cell cycle during germ cell formation is-

(A). Meiosis I anaphase

(B). Meiosis II anaphase

(C). G1

(D). G2

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)

2. (A), (C), (B), (D)

3. (C), (D), (A), (B)

4. (C), (B), (D), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.41 Read the following passage carefully and answer the given questions.

Biofertilisers

Biofertilisers are organisms that enrich the nutrient quality of the soil and their main sources are bacteria, fungi and cyanobacteria. You have studied about the nodules on the roots of leguminous plants formed by the symbiotic association of *Rhizobium*. These bacteria can fix atmospheric nitrogen into organic forms. Other bacteria can also fix atmospheric nitrogen while free-living in the soil (examples *Azospirillum* and *Azotobacter*). Fungi are also known to form symbiotic associations with plants (mycorrhiza). Many members of the genus *Glomus* form mycorrhiza. Plants having such associations show other benefits also, such as resistance to root-borne pathogens, tolerance to salinity and drought, and an overall increase in plant growth and development. Cyanobacteria are autotrophic microbes widely distributed in aquatic and terrestrial environments many of which can fix atmospheric nitrogen, e.g. *Anabaena*, *Nostoc*, *Oscillatoria*, etc. In paddy fields, cyanobacteria serve as an important biofertiliser. Blue green algae also add organic matter in to the soil and increase its fertility. Currently, in our country, a number of biofertilisers are available commercially in the market and farmers use these regularly in their fields to replenish soil nutrients and to reduce dependence on chemical fertilisers.

In organic farming, there is -

1. Overuse of chemical fertilisers

2. Use of biofertilisers

3. No use of fertilisers

4. Only use of symbiotic bacteria

Options 1. 1

2. 2

3. 3

4. 4

Q.42 Read the following passage carefully and answer the given questions.

Biofertilisers

Biofertilisers are organisms that enrich the nutrient quality of the soil and their main sources are bacteria, fungi and cyanobacteria. You have studied about the nodules on the roots of leguminous plants formed by the symbiotic association of *Rhizobium*. These bacteria can fix atmospheric nitrogen into organic forms. Other bacteria can also fix atmospheric nitrogen while free-living in the soil (examples *Azospirillum* and *Azotobacter*). Fungi are also known to form symbiotic associations with plants (mycorrhiza). Many members of the genus *Glomus* form mycorrhiza. Plants having such associations show other benefits also, such as resistance to root-borne pathogens, tolerance to salinity and drought, and an overall increase in plant growth and development. Cyanobacteria are autotrophic microbes widely distributed in aquatic and terrestrial environments many of which can fix atmospheric nitrogen, e.g. *Anabaena*, *Nostoc*, *Oscillatoria*, etc. In paddy fields, cyanobacteria serve as an important biofertiliser. Blue green algae also add organic matter in to the

soil and increase its fertility. Currently, in our country, a number of biofertilisers are available commercially in the market and farmers use these regularly in their fields to replenish soil nutrients and to reduce dependence on chemical fertilisers.

Choose the free living, nitrogen-fixing bacteria from the following.

1. Glomus
2. *Rhizobium*
3. *Azospirillum*
4. Mycorrhiza

Options 1. 1

2. 2
3. 3
4. 4

Q.43 Read the following passage carefully and answer the given questions.

Biofertilisers

Biofertilisers are organisms that enrich the nutrient quality of the soil and their main sources are bacteria, fungi and cyanobacteria. You have studied about the nodules on the roots of leguminous plants formed by the symbiotic association of *Rhizobium*. These bacteria can fix atmospheric nitrogen into organic forms. Other bacteria can also fix atmospheric nitrogen while free-living in the soil (examples *Azospirillum* and *Azotobacter*). Fungi are also known to form symbiotic associations with plants (mycorrhiza). Many members of the genus *Glomus* form mycorrhiza. Plants having such associations show other benefits also, such as resistance to root-borne pathogens, tolerance to salinity and drought, and an overall increase in plant growth and development. Cyanobacteria are autotrophic microbes widely distributed in aquatic and terrestrial environments many of which can fix atmospheric nitrogen, e.g. *Anabaena*, *Nostoc*, *Oscillatoria*, etc. In paddy fields, cyanobacteria serve as an important biofertiliser. Blue green algae also add organic matter in to the soil and increase its fertility. Currently, in our country, a number of biofertilisers are available commercially in the market and farmers use these regularly in their fields to replenish soil nutrients and to reduce dependence on chemical fertilisers.

Which one of the followings is not a symbiotic microbe?

1. *Rhizobium*
2. Mycorrhiza
3. *Azotobacter*
4. Lichens

Options 1. 1

2. 2
3. 3
4. 4

Q.44 Read the following passage carefully and answer the given questions.

Biofertilisers

Biofertilisers are organisms that enrich the nutrient quality of the soil and their main sources are bacteria, fungi and cyanobacteria. You have studied about the nodules on the roots of leguminous plants formed by the symbiotic association of *Rhizobium*. These bacteria can fix atmospheric nitrogen into organic forms. Other bacteria can also fix atmospheric nitrogen while free-living in the soil (examples *Azospirillum* and *Azotobacter*). Fungi are also known to form symbiotic associations with plants (mycorrhiza). Many members of the genus *Glomus* form mycorrhiza. Plants having such associations show other benefits also, such as resistance to root-borne pathogens, tolerance to salinity and drought, and an overall increase in plant growth and development. Cyanobacteria are autotrophic microbes widely distributed in aquatic and terrestrial environments many of which can fix atmospheric nitrogen, e.g. *Anabaena*, *Nostoc*, *Oscillatoria*, etc. In paddy fields, cyanobacteria serve as an important biofertiliser. Blue green algae also add organic matter in to the soil and increase its fertility. Currently, in our country, a number of biofertilisers are available commercially in the market and farmers use these regularly in their fields to replenish soil nutrients and to reduce dependence on chemical fertilisers.

Resistance to root-borne pathogens, salinity and drought tolerance are introduced into which type of plants?

1. Xerophytic plants

2. Terrestrial plants
3. Plants having symbiotic relations with fungi
4. Non-legume plants

Options 1. 1

2. 2
3. 3
4. 4

Q.45 Read the following passage carefully and answer the given questions.

Biofertilisers

Biofertilisers are organisms that enrich the nutrient quality of the soil and their main sources are bacteria, fungi and cyanobacteria. You have studied about the nodules on the roots of leguminous plants formed by the symbiotic association of *Rhizobium*. These bacteria can fix atmospheric nitrogen into organic forms. Other bacteria can also fix atmospheric nitrogen while free-living in the soil (examples *Azospirillum* and *Azotobacter*). Fungi are also known to form symbiotic associations with plants (mycorrhiza). Many members of the genus *Glomus* form mycorrhiza. Plants having such associations show other benefits also, such as resistance to root-borne pathogens, tolerance to salinity and drought, and an overall increase in plant growth and development. Cyanobacteria are autotrophic microbes widely distributed in aquatic and terrestrial environments many of which can fix atmospheric nitrogen, e.g. *Anabaena*, *Nostoc*, *Oscillatoria*, etc. In paddy fields, cyanobacteria serve as an important biofertiliser. Blue green algae also add organic matter in to the soil and increase its fertility. Currently, in our country, a number of biofertilisers are available commercially in the market and farmers use these regularly in their fields to replenish soil nutrients and to reduce dependence on chemical fertilisers.

Nostoc is an example of :

1. *Glomus* family
2. Autotrophic microbe
3. Heterotrophic microbe
4. Symbiotic fungi

Options 1. 1

2. 2
3. 3
4. 4

Q.46 Read the following passage carefully and answer the given questions.

Immune System in the Body

The human immune system consists of lymphoid organs, tissues, cells and soluble molecules like antibodies. The primary lymphoid organs are bone marrow and thymus where immature lymphocytes differentiate into antigen-sensitive lymphocytes. After maturation the lymphocytes migrate to secondary lymphoid organs like spleen, lymph nodes, tonsils, Peyer's patches of small intestine and appendix. The secondary lymphoid organs provide the sites for interaction of lymphocytes with the antigen, which then proliferate to become effector cells. The bone marrow is the main lymphoid organ where all blood cells including lymphocytes are produced. Each antibody molecule has four peptide chains, two small called light chains and two longer called heavy chains. Hence, an antibody is represented as H₂L₂. Different types of antibodies are produced in our body. IgA, IgM, IgE, IgG are some of them. Techniques like radiography (use of X-rays), CT (computed tomography) and MRI (magnetic resonance imaging) are very useful to detect cancers of the internal organs.

The site of interaction of lymphocytes with antigen is -

1. Pancreas
2. Secondary lymphoid organs
3. Bone marrow
4. Thymus

Options 1. 1

2. 2
3. 3
4. 4

Q.47 Read the following passage carefully and answer the given questions.

Immune System in the Body

The human immune system consists of lymphoid organs, tissues, cells and soluble molecules like antibodies. The primary lymphoid organs are bone marrow and thymus where immature lymphocytes differentiate into antigen-sensitive lymphocytes. After maturation the lymphocytes migrate to secondary lymphoid organs like spleen, lymph nodes, tonsils, Peyer's patches of small intestine and appendix. The secondary lymphoid organs provide the sites for interaction of lymphocytes with the antigen, which then proliferate to become effector cells. The bone marrow is the main lymphoid organ where all blood cells including lymphocytes are produced. Each antibody molecule has four peptide chains, two small called light chains and two longer called heavy chains. Hence, an antibody is represented as H2L2. Different types of antibodies are produced in our body. IgA, IgM, IgE, IgG are some of them. Techniques like radiography (use of X-rays), CT (computed tomography) and MRI (magnetic resonance imaging) are very useful to detect cancers of the internal organs.

Which one of the following technique is not useful for the detection of cancers of internal organs?

1. Radiography
2. CT
3. DNA fingerprinting
4. MRI

Options 1. 1

2. 2
3. 3
4. 4

Q.48 Read the following passage carefully and answer the given questions.

Immune System in the Body

The human immune system consists of lymphoid organs, tissues, cells and soluble molecules like antibodies. The primary lymphoid organs are bone marrow and thymus where immature lymphocytes differentiate into antigen-sensitive lymphocytes. After maturation the lymphocytes migrate to secondary lymphoid organs like spleen, lymph nodes, tonsils, Peyer's patches of small intestine and appendix. The secondary lymphoid organs provide the sites for interaction of lymphocytes with the antigen, which then proliferate to become effector cells. The bone marrow is the main lymphoid organ where all blood cells including lymphocytes are produced. Each antibody molecule has four peptide chains, two small called light chains and two longer called heavy chains. Hence, an antibody is represented as H2L2. Different types of antibodies are produced in our body. IgA, IgM, IgE, IgG are some of them. Techniques like radiography (use of X-rays), CT (computed tomography) and MRI (magnetic resonance imaging) are very useful to detect cancers of the internal organs.

An antibody is represented as -

1. H1L3
2. H1L2
3. H2L2
4. H2L1

Options 1. 1

2. 2
3. 3
4. 4

Q.49 Read the following passage carefully and answer the given questions.

Immune System in the Body

The human immune system consists of lymphoid organs, tissues, cells and soluble molecules like antibodies. The primary lymphoid organs are bone marrow and thymus where immature lymphocytes differentiate into antigen-sensitive lymphocytes. After maturation the lymphocytes migrate to secondary lymphoid organs like spleen, lymph nodes, tonsils, Peyer's patches of small intestine and appendix. The secondary lymphoid organs provide the sites for interaction of lymphocytes with the antigen, which then proliferate to become effector cells. The bone marrow is the main lymphoid organ where all blood cells including lymphocytes are produced. Each antibody molecule has four peptide chains, two small called light chains and two longer called heavy chains. Hence, an antibody is represented as H2L2. Different types of antibodies are produced in our

body. IgA, IgM, IgE, IgG are some of them. Techniques like radiography (use of X-rays), CT (computed tomography) and MRI (magnetic resonance imaging) are very useful to detect cancers of the internal organs.

Maturation of lymphocytes takes place in:

1. Secondary lymphoid organs
2. Bone marrow only
3. Bone marrow and thymus
4. Lymph nodes

Options 1. 1

2. 2

3. 3

4. 4

Q.50 Read the following passage carefully and answer the given questions.

Immune System in the Body

The human immune system consists of lymphoid organs, tissues, cells and soluble molecules like antibodies. The primary lymphoid organs are bone marrow and thymus where immature lymphocytes differentiate into antigen-sensitive lymphocytes. After maturation the lymphocytes migrate to secondary lymphoid organs like spleen, lymph nodes, tonsils, Peyer's patches of small intestine and appendix. The secondary lymphoid organs provide the sites for interaction of lymphocytes with the antigen, which then proliferate to become effector cells. The bone marrow is the main lymphoid organ where all blood cells including lymphocytes are produced. Each antibody molecule has four peptide chains, two small called light chains and two longer called heavy chains. Hence, an antibody is represented as H₂L₂. Different types of antibodies are produced in our body. IgA, IgM, IgE, IgG are some of them. Techniques like radiography (use of X-rays), CT (computed tomography) and MRI (magnetic resonance imaging) are very useful to detect cancers of the internal organs.

Human immune system recognizes the viral proteins as:

1. Antibody
2. Antibiotic
3. Antigen
4. Soluble immune molecule

Options 1. 1

2. 2

3. 3

4. 4